

PAPER_846: Chandra Death Star BHs — 16 SMBH Composite, Abell 478, NGC 5044, GC Vent, Cas A + Crab Timelapse

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Abstract

Chandra Batch 4 examines five system groups: Death Star BHs (16 SMBHs with destructive jets), Abell 478 (galaxy cluster), NGC 5044 (elliptical galaxy group), Galactic Center Vent (bipolar outflow), and Cassiopeia A + Crab Nebula timelapse evolution. The GC Vent shares Sgr A*'s parameters and exhibits the same negative buoyancy ($F_{U_{Bi}} \sim -8.31e211$ N), confirming that bipolar outflow structures inherit the SMBH's gravitational scale reversal.

1. Death Star BHs — 16 SMBH Composite

16 galaxies where SMBH jets destroy companion gas/stars
"Death Star" analogy: focused relativistic jets ablate neighbors

Composite: $M = 1.59e40$ kg, $r = 3.09e22$ m

Jet kinetic power:

$$P_{jet} = 0.5 * M_{dot} * (v_{jet} * c)^2$$

$M_{dot} = 10^{20}$ kg/s (accretion rate)

$v_{jet} = 0.1c$ (relativistic jet)

$$P_{jet} = 0.5 * 10^{20} * (0.1 * 3e8)^2 = 4.49e34$$
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2. Galaxy Cluster and Group

System	M (kg)	r (m)	$F_{U_{Bi}}$
Abell 478	$1e45$	$1.85e23$	$\sim 2.65e208$ N (positive)
NGC 5044	$3e42$	$6.17e22$	$\sim 2.65e208$ N (positive)

Both are relaxed systems with cool cores.
SMBH feedback regulates cooling flows.

3. Galactic Center Vent

Bipolar chimney structure extending ~ 500 pc from Sgr A*
Same M and r as Galactic Center:

$M = 7.956e36$ kg, $r = 6.17e18$ m

$F_{U_{Bi}} \sim -8.31e211$ N (NEGATIVE BUOYANCY)

The vent inherits Sgr A*'s gravitational parameters.

Negative buoyancy in the vent may drive the observed bipolar outflow structure (hot gas rising against gravitational attraction creates vacuum-mediated chimney).

4. Cas A + Crab Timelapse

Temporal evolution study spanning 20+ years of Chandra observations.

Cas A: expansion velocity ~5000 km/s, age ~340 years

Crab: expansion velocity ~1500 km/s, age ~970 years

UQFF $F_{U_{Bi}}$ is approximately constant over these timescales
(M and r evolve slowly compared to observation baseline).

Conclusion

The Death Star BH composite demonstrates SMBH jet feedback across 16 systems. The GC Vent's negative buoyancy ($-8.31e211$ N) confirms that bipolar outflow structures inherit their parent SMBH's gravitational scale reversal, providing a physical mechanism for chimney formation.

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